SQL Data Analysis — Report

Project 3 · Data Analytics Internship · Decode Labs

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Position:	Data Analytics Intern	**Mode:**	Remote / Virtual
Internship Period:	27 Sep 2026 – 27 Oct 2026	**Report Date:**	14 Oct 2026
Dataset:	Orders_Clean (1,200 rows)	**Tool Used:**	SQLite (DB Browser)

Objective

Use SQL queries to extract insights from a dataset, per the Project 3 brief: write SELECT queries, use WHERE, ORDER BY and GROUP BY, and perform basic aggregations (COUNT, SUM, AVG). The analysis runs on the same cleaned order data used in Projects 1 and 2, so SQL results can be checked directly against the earlier Excel work.

Setup

The cleaned Orders_Clean sheet from Project 1 was exported from Excel as a CSV file and imported into a SQLite database (Orders_Database.db) using DB Browser for SQLite, with the first row used as column names. The import was verified with SELECT COUNT(*), which returned 1,200 rows, matching the Project 1 row count. Quantity and ItemsInCart were imported as INTEGER, UnitPrice and TotalPrice as REAL, and all other columns as TEXT.

1. Query Log

#	Purpose	SQL used	Result
1	Preview the first 10 rows	SELECT, LIMIT	10 rows, 14 columns
2	Count all rows	COUNT	1,200
3	Select specific columns	SELECT, LIMIT	4 columns, 10 rows
4	Filter by product	WHERE	Laptop orders only
5	Count Laptop orders	WHERE, COUNT	173
6	Ten highest-value orders	ORDER BY DESC, LIMIT	10 rows
7	Orders above outlier cutoff (3,330.41)	WHERE, COUNT	8
8	Laptop orders of exactly 5 units	WHERE, AND, ORDER BY	39 rows
9	Orders and average value per product	GROUP BY, COUNT, AVG	7 rows
10	Orders and average value per referral source	GROUP BY, COUNT, AVG	5 rows
11	Orders and total revenue per product	GROUP BY, COUNT, SUM	7 rows
12	Products with more than 170 orders	GROUP BY, HAVING	4 rows
13	Revenue share per product	GROUP BY, SUM, subquery	7 rows

The full, commented SQL for all 13 queries is in Project3_SQL_Queries.sql.

2. Key Queries and Results

Query 9 — Orders and average order value per product (GROUP BY, COUNT, AVG)

```
SELECT Product,
       COUNT(*) AS Order_Count,
       ROUND(AVG(TotalPrice), 2) AS Avg_TotalPrice
FROM Orders_Clean
GROUP BY Product
ORDER BY Avg_TotalPrice DESC;
```

Product	Order_Count	Avg_TotalPrice
Laptop	173	1,110.56
Chair	178	1,098.99
Printer	181	1,080.73
Monitor	163	1,077.62
Tablet	179	1,042.28
Desk	170	985.06
Phone	156	972.58

Query 10 — Orders and average order value per referral source

ReferralSource	Order_Count	Avg_TotalPrice
Facebook	228	1,098.29
Instagram	259	1,062.88
Email	250	1,047.23
Google	241	1,039.18
Referral	222	1,021.69

Query 11 — Total revenue per product (SUM) and Query 13 — revenue share

```
SELECT Product,
       ROUND(SUM(TotalPrice), 2) AS Total_Revenue,
       ROUND(100.0 * SUM(TotalPrice) /
             (SELECT SUM(TotalPrice) FROM Orders_Clean), 2) AS Revenue_Share_Pct
FROM Orders_Clean
GROUP BY Product
ORDER BY Total_Revenue DESC;
```

Product	Order_Count	Total_Revenue	Revenue_Share_Pct
Chair	178	195,620.11	15.47
Printer	181	195,612.61	15.47
Laptop	173	192,126.56	15.19
Tablet	179	186,568.95	14.75
Monitor	163	175,651.41	13.89
Desk	170	167,459.93	13.24
Phone	156	151,722.39	12.00

Order counts come from Query 11; shares are from Query 13 (they sum to 100.01 because of rounding).

Query 12 — Filtering groups with HAVING

```
SELECT Product,
       COUNT(*) AS Order_Count
FROM Orders_Clean
GROUP BY Product
HAVING COUNT(*) > 170
ORDER BY Order_Count DESC;
```

Product	Order_Count
Printer	181
Tablet	179
Chair	178
Laptop	173

HAVING filters whole groups after aggregation, whereas WHERE filters individual rows before grouping. Desk has exactly 170 orders, so it is correctly excluded by the strict > 170 condition.

3. Cross-Validation Against Earlier Excel Work

Check	SQL result	Excel result (Projects 1-2)	Match
Total rows	1,200	1,200 (Project 1)	Yes
Laptop order count	173	173 (Product PivotTable)	Yes
Orders above 3,330.41	8	8 (IQR outlier count)	Yes

Check	SQL result	Excel result (Projects 1-2)	Match
Product counts and averages	Query 9	Product PivotTable	Yes
Referral counts and averages	Query 10	Referral PivotTable	Yes
Sum of product revenues	1,264,761.96	1,200 x 1,053.9683 = 1,264,761.96	Yes

Query 8 (39 Laptop orders of 5 units) has no equivalent earlier result to compare against.

4. Key Findings

1. Laptop has the highest average order value (1,110.56) but ranks third in total revenue (192,126.56), because it has fewer orders (173) than Chair (178) and Printer (181). Value per order and total revenue give different rankings.

2. Chair and Printer are effectively tied for the highest revenue: 195,620.11 versus 195,612.61, a gap of 7.50. Each contributes about 15.47% of total revenue.

3. Phone is last on every measure: fewest orders (156), lowest average order value (972.58), and lowest revenue (151,722.39, a 12.00% share). This is consistent with the Project 2 finding.

4. Revenue is spread fairly evenly across products. Shares range from 12.00% to 15.47%, so no single product dominates the business.

5. Among referral channels, Facebook has the highest average order value (1,098.29) while Instagram brings the most orders (259).

Notes

The dataset does not state a currency, so monetary values are shown as plain numbers. All queries were run in DB Browser for SQLite, one at a time, against the Orders_Clean table.

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